

Mercurial

A Distributed Version Control System (DVCS)

This article is meant to introduce you to the concept of a DVCS. It is followed by a review of the Mercurial (hg) DVCS and a short comparison with Git. The author assumes that you know the core concepts of version control. The article will not attempt to teach you how to use Mercurial in two pages, but what it will do is point you to a couple of really good tutorials out there. :)

Version control systems have always been varied in architecture and implementation (on the basis of both language and algorithms) across different platforms. Let's start off by discussing a few key characteristics of a DVCS.

DVCS characteristics

- Being *distributed* means that you have no central location for tracking data. No single location (repository) is more important than any other.
- You could take your machine offline and still make commits, check logs, create *diffs*, etc. This is very important when you have hundreds of developers working on a project, from different parts of the globe. A key feature of a truly distributed system is that it allows you to do most of the work without accessing a central server.
- The biggest difference between a client/server VCS and a DVCS is that branches are used a lot more. In the former, branches are mostly used when you wish to do some long-term development. The latter provides a nice level of isolation between the disparate features you might be working on. Even if you consider your project as having only a single branch, the code on your local machine can be viewed as a separate branch every time you disconnect from the Internet.

- As a consequence of branching being more inherent in DVCSs (it's a cheap operation, compared to non-distributed version control systems), developers usually have a fair number of branches for themselves! For example: one for maintenance, one for some experimental work, one for a new feature addition, etc.

Mercurial

Decentralisation, high performance, scalability, robust handling of text and binary files while remaining conceptually simple—these are a few design goals with which hg or Mercurial has been designed. It is a platform-independent, distributed revision control tool for software developers, written in the Python programming language. Mercurial was initially written to run on Linux; now it has been ported to Windows, MacOS X, and most UNIX-like systems. It is primarily a command-line program, but graphical user interface extensions are also available. Of course, it's an open source project, and if you are interested in contributing, you can find information about the mailing lists, subscription and archive links, at <http://mercurial.selenic.com/wiki/MailingLists>.

The promised tutorials

Here are a few recommended tutorials:

<http://mercurial.selenic.com/wiki/Tutorial>

This is the standard documentation, which is available at Mercurial's home page. It's very